

A Low-Frequency Momentum Trading Strategy on Turkish Equities (BIST100)

EC581 – Algorithmic Trading and Quantitative Strategies
Project Section 4: Trend Following

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Abstract

We design, implement and evaluate a long/short trend-following strategy for Turkish equities (BIST100) inspired by Harris and Yilmaz (2008), *A Momentum Trading Strategy Based on the Low Frequency Component of the Exchange Rate*. Four trend extractors — EMA crossover (S1), EMA direction (S2), Hodrick–Prescott filter direction (S3), and LOWESS direction (S4) — are tuned in-sample on the BIST100 Index (Phase 1) and then deployed on 29 constituent stocks (Phase 2) with an optional BIST100-HP regime gate. Statistical significance is established with a run-length-matched Monte-Carlo bootstrap ($N=1000$). The Hodrick–Prescott direction rule (S3) is retained as the principled primary strategy by virtue of its theoretical alignment with Harris–Yilmaz; LOWESS (S4) is carried as the *empirical* benchmark and reported head-to-head. On the BIST100 Index over 23.3 years, S3 HP reaches Sharpe 0.82 ($p < 0.001$ vs. matched-random) and S4 LOWESS reaches Sharpe 0.88 ($p < 0.001$). The equal-weight 29-stock portfolio gives S3 HP a Sharpe of 0.90 (CAGR 15.3%, MDD -26.8%) and S4 LOWESS a Sharpe of 1.05 (CAGR 18.2%, MDD -28.5%), compared to BIST100 buy-and-hold Sharpe 0.92 (CAGR 23.5%, MDD -63.4%). The regime filter consistently acts as a *variance reducer* rather than a Sharpe enhancer. We close with a detailed methodological self-critique and a set of concrete improvement directions covering universe construction, signal combination, position sizing, transaction-cost realism, and statistical testing.

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1 Introduction

1.1 Project Scope

This report documents the project for Section 4 (*Trend Following*) of the EC581 course brief at Bogazici University. The anchoring article is Harris and Yilmaz (2008/2009), *A Momentum Trading Strategy Based on the Low Frequency Component of the Exchange Rate*, published in the *Journal of Banking and Finance*. The brief mandates the following constraints, all of which are respected:

- Backtest engine: **Backtrader**.
- Initial capital: **1,000,000 TL**.
- Per-trade notional: **100,000 TL**, allocated via the course-supplied **FixedCashSizer**.
- Commissions and slippage: **zero**.
- Order type (base strategy): **bt.Order.Market**.
- Frequency: **daily** bars.
- Universe Phase 1: **BIST100 Index** for hyperparameter selection.
- Universe Phase 2: **BIST100 constituents** (current membership; survivorship bias acknowledged but uncorrected per brief).
- Significance: **Monte-Carlo p -value** on the Sharpe ratio against random strategies matching trade frequency, holding period and long/short proportion, with $N \approx 1000$.

The project is to (i) implement the four trend extractors described in §3, (ii) tune each on the index, (iii) pick a primary, (iv) deploy it on individual stocks both in a base and a regime-gated form, and (v) test statistical significance throughout.

1.2 Why Trend Following?

Classical efficient-market theory predicts no exploitable serial dependence in returns. In practice, persistent autocorrelation in returns has been documented across asset classes and decades (Moskowitz, Ooi and Pedersen, 2012; Asness, Moskowitz and Pedersen, 2013). Behavioural explanations — anchoring, under-reaction to news, herding — and structural ones — slow capital flows, leverage cycles, central-bank reaction functions — all predict that prices adjust to information gradually rather than instantaneously. A strategy that is *long when the slow trend is up and short when it is down* aims to harvest that drift.

1.3 Why Low-Frequency Filters?

Harris and Yilmaz (2008) decompose daily FX series into a low-frequency component (extracted with the Hodrick–Prescott filter from business-cycle econometrics) and a high-frequency residual. Their three operationally useful claims are:

1. The low-frequency component is *highly persistent*: once its slope flips sign, it tends to keep that sign for many periods.
2. Rules based on the **sign of the change in the low-frequency component** outperform classical moving-average crossover rules out-of-sample across major currency pairs.
3. Low-pass filters, while non-causal during fitting, produce smoother right-edge estimates than moving averages of comparable lag, reducing whipsaws.

We test these claims on Turkish equities.

2 Theoretical Background

2.1 The Hodrick–Prescott Filter

Given a price series $\{P_t\}_{t=1}^T$, the HP decomposition writes $P_t = \tau_t + c_t$, where τ_t is a smooth trend and c_t the residual cycle. The trend solves

$$\hat{\tau} = \arg \min_{\tau} \underbrace{\sum_{t=1}^T (P_t - \tau_t)^2}_{\text{fit}} + \lambda \underbrace{\sum_{t=2}^{T-1} ((\tau_{t+1} - \tau_t) - (\tau_t - \tau_{t-1}))^2}_{\text{smoothness}}, \quad (1)$$

with closed form

$$\hat{\tau} = (\mathbf{I} + \lambda \mathbf{D}^\top \mathbf{D})^{-1} \mathbf{P}, \quad (2)$$

where $\mathbf{D}$ is the second-difference operator. Larger λ implies a smoother (slower) trend. Hodrick and Prescott (1997) recommend $\lambda=1600$ for quarterly macroeconomic data; on daily financial series much larger values (e.g. 10^4 – 10^5) are typical.

Causality caveat. Equation (2) uses the *entire* series. In a backtest at bar t , the historical estimate $\hat{\tau}_t^{\text{full}}$ depends on future prices and is therefore *look-ahead biased*. We address this with a rolling/expanding-window wrapper (Section 4.2).

2.2 LOWESS Smoothing

LOWESS (Cleveland, 1979) fits a low-degree polynomial via locally weighted least-squares around each target abscissa x_t . The weighting uses a tricube kernel over the nearest $\text{frac} \cdot T$ neighbours. As with HP, the bare smoother is non-causal in the same global sense: each smoothed point uses neighbours on both sides. The same rolling-window discipline applies.

2.3 Exponential Moving Average

The EMA of span n is recursively

$$\text{EMA}_t = \alpha P_t + (1 - \alpha) \text{EMA}_{t-1}, \quad \alpha = \frac{2}{n+1}. \quad (3)$$

The EMA is *causal by construction*: EMA_t only depends on $\{P_s\}_{s \leq t}$. It is therefore the natural baseline.

3 Strategies

Each strategy produces a binary trend state $T_t \in \{-1, +1\}$ on a daily bar.

#	Name	Indicator	Tunable parameter(s)
S1	EMA Crossover	$\text{sign}(\text{EMA}_{n_f} - \text{EMA}_{n_s})$	(n_f, n_s)
S2	EMA Direction	$\text{sign}(\Delta \text{EMA}_n)$	n
S3	HP Direction (<i>primary</i>)	$\text{sign}(\Delta \text{HP}_\lambda(P))$	λ , window W
S4	LOWESS Direction	$\text{sign}(\Delta \text{LOWESS}_{\text{frac}}(P))$	frac , window W

Table 1: Four trend extractors. All produce a discrete ± 1 trend state.

Trade rules (common to all variants).

- **Long entry** when T_t flips from -1 to $+1$.
- **Close long and short entry** when T_t flips from $+1$ to -1 .
- **Always in market:** no flat state in the base strategy.

Position sizing — full capital deployment. Sizing uses the course-mandated `FixedCashSizer` which targets $\sim 100,000$ TL of notional per individual `buy()`/`sell()` order. A naive implementation that holds a single 100k lot at a time would leave 90% of the 1M TL capital idle, producing misleading single-digit CAGRs. We therefore fire $\lfloor \text{equity}/100,000 \rfloor$ market orders on each signal flip (≈ 10 orders for a fresh 1M account), close any opposite position first, and hold the stacked lots until the next flip. This respects the per-trade cap while keeping capital fully invested.

Why HP is designated the primary strategy. Per Section 2, the HP-direction rule maps directly onto the Harris–Yilmaz argument that a low-frequency filter’s directional sign is a clean trend signal. Empirically, LOWESS edges HP at every aggregation level (Section 5.7); the difference is small enough that retaining HP as the principled primary and reporting LOWESS as the head-to-head benchmark is honest reporting rather than benchmark-swapping.

3.1 Regime Filter (Phase 2)

After picking a primary on the index, we deploy it on individual stocks *gated by the BIST100’s own HP-trend*:

- **Long entry** on stock i requires: stock signal long **AND** BIST100 close $>$ HP-filtered BIST100 close.
- **Short entry** on stock i requires: stock signal short **AND** BIST100 close $<$ HP-filtered BIST100 close.
- **Exits** ignore the regime filter (to avoid trapping positions when the regime flips intra-trade).

The economic story is straightforward top-down filtering: trade with the index, not against it.

4 Methodology**4.1 Data and Universe**

- **Phase 1 universe:** BIST100 Index daily close, ticker `XU100.IS` on Yahoo Finance.
- **Phase 2 universe:** 29 BIST100 constituents (cleaned from a curated list of 31 well-known blue chips, two failing yfinance ingest as “possibly delisted”). Three additional placeholders (`SODA.IS`, `KOZAL.IS`, `KOZAA.IS`) need ticker remapping before they can be re-added. Survivorship bias is acknowledged but not corrected, per brief.
- **Data fields:** OHLCV in TL, daily frequency.
- **Period:** 2003-01-01 to 2026-05-11 (~ 23 years). Panel shape: 5997×29 . PGSUS has the shortest history at 3273 bars (later IPO); all 29 clear the `MIN_HISTORY_DAYS=750` floor.

Currency redenomination and corporate-action fix. `yfinance` does not back-adjust for the 2005-01-03 Turkish TL 1M-to-1 redenomination, leaving a $\sim 1000\times$ single-day price jump in every Turkish equity series. The function `src/data/clean.py::_adjust_splits` detects any single-day price ratio $> 3\times$ (or $< 1/3\times$) and rescales all prior OHLC values by that ratio, idempotently. The $3\times$ threshold (originally $50\times$) also catches the MGROS 2009-08-04 corporate-action gap (factor $4.04\times$) that previously caused -14358% MDD blowups on shorts. The largest legitimate single-day move in the universe is HEKTS $+44\%$, well below the threshold.

4.2 Causal Wrappers for HP and LOWESS

The brief's `hp_filter(x, lam=1600)` and the `statsmodels.lowess(...)` routines both operate on the *whole* input series at once. Used directly inside a backtest, they leak future data into past trend values. We therefore wrap both in rolling-window helpers (`src/features/hp.py::rolling_hp_trend` and `src/features/lowess.py::rolling_lowess_trend`) that, at each bar t , re-fit on the trailing window $[t - W, t]$ and return only the rightmost trend value. The bare functions remain available for EDA and plotting; the backtest pipeline never sees them.

4.3 Backtest Mechanics

A single Backtrader Cerebro instance per (strategy, ticker) pair keeps P&L attribution clean. The wiring is:

- **Feed** (`src/backtest/feeds.py`): `PandasData` extended with a precomputed `trend` line and (for the regime variant) a `regime` line.
- **Strategy** (`src/backtest/strategies.py`): reads `self.data.trend[0]`, acts on sign flips, sizes orders with `FixedCashSizer(100_000)`.
- **Runner** (`src/backtest/runner.py`): orchestrates the Cerebro lifecycle, attaches analyzers for per-bar equity and per-trade P&L, returns a structured `RunResult`.

Backtrader's `next()` with market orders fills the next bar; signals generated on bar t therefore translate to fills at $t + 1$, which is the correct convention.

4.4 Performance Metrics

Reported for every (strategy, variant, ticker) combination:

- | | |
|---------------------------------------|---|
| 1. Net P/L (TL and %) | 6. Max drawdown and drawdown duration |
| 2. Compound Annual Growth Rate (CAGR) | 7. Calmar ratio |
| 3. Annualised volatility | 8. Win rate, average win, average loss |
| 4. Sharpe ratio (annualised) | 9. Number of trades, average holding period |
| 5. Sortino ratio (annualised) | 10. Turnover |

4.5 Hyperparameter Optimisation

Phase 1 grid (BIST100 Index) via `src/backtest/sweep.py::default_specs()`:

- **S1:** $n_f \in \{5, 10, 20, 30\}$, $n_s \in \{50, 100, 200\}$ with $n_f < n_s$.
- **S2:** $n \in \{10, 20, 50, 100, 200\}$.
- **S3:** $\lambda \in \{100, 1600, 14400, 129600\}$, window $W \in \{252, 504, 1260\}$.
- **S4:** $\text{frac} \in \{0.05, 0.1, 0.2, 0.3\}$, window same as S3.

Selection rule: highest Sharpe, with a minimum-trade floor (≥ 30 round trips) to disqualify degenerate configurations.

Phase 2 walk-forward: 3-year training, 1-year test, 1-year step, with parameter re-selection on each training window. The 5-year ($W = 1260$) configurations are pruned because their warmup leaves too little post-warmup room for fold-level retraining.

4.6 Statistical Significance: Monte-Carlo Bootstrap

For each (strategy, ticker) we generate $N = 1000$ random binary signals that match the empirical:

1. run-length distribution (a proxy for trade frequency $\times$ holding period);
2. long/short proportion.

Each random signal is then scored on the same prices with the same simulator, producing a null distribution of Sharpe ratios. The one-sided p -value is the fraction of random Sharpes exceeding the strategy's. The fast vectorised simulator in `src/eval/montecarlo.py` is used for both the strategy and the matched-random pool to keep the comparison apples-to-apples.

4.7 Portfolio Aggregation

The equal-weight cross-sectional portfolio (DESIGN.md milestone M4) is built by an `effective_position` state machine in `src/eval/portfolio.py` that mirrors `TrendStrategy` and `TrendRegimeStrategy` exactly, including the regime variant's *sticky-flat* entry behaviour. Per ticker, this gives a $\{-1, 0, +1\}$ held-position series. The portfolio return on bar t is the cross-sectional mean of per-ticker strategy returns ($\text{pos}_{t-1} \cdot \Delta \ln P_t$), skipping pre-IPO NaNs and counting flat-but-active tickers as zero contributions.

5 Results

5.1 Phase 1 – BIST100 Index Sweep

The 41-configuration grid (run time ≈ 17 min on a single core, full deployment) produces the following best-by-Sharpe winners (`results/phase1_sweep.parquet`):

Strategy	Best params	Sharpe	Trades	CAGR	MDD
S4 LOWESS direction	frac=0.2, $W=252$	0.876	560	19.89%	−46.4%
S3 HP direction (<i>primary</i>)	$\lambda=14400$, $W=504$	0.818	576	17.52%	−43.3%
S2 EMA direction	$n=20$	0.742	634	16.71%	−47.1%
S1 EMA crossover	$n_f=5$, $n_s=100$	0.719	126	15.60%	−52.8%
<i>Reference: BIST100 buy-and-hold (23.3y)</i>		0.92	—	23.46%	−63.4%

Table 2: Phase-1 winners on the BIST100 Index. The strategies underperform buy-and-hold on absolute return but cut drawdown by 15–20 percentage points.

Reading. Four observations stand out:

1. All four strategies are competitive with buy-and-hold in Sharpe terms.
2. LOWESS and HP — the two low-pass filters — beat the two EMA strategies, consistent with the Harris–Yilmaz hypothesis.

3. S1 (the classical MA crossover) is the weakest, with the fewest trades. Crossover rules wait for a magnitude-difference rather than a slope-change, so they enter later and exit later.
4. Drawdowns are 15–20pp shallower than buy-and-hold but still ~45–53%. This is the cost of always-in-market with shorts.

5.2 Phase 1 – Monte-Carlo Significance

Strategy	Sharpe	MC mean	MC q_{95}	p -value
S4 LOWESS	0.87	0.16	0.50	< 0.001
S2 EMA dir	0.86	0.19	0.54	0.002
S3 HP	0.80	0.14	0.47	0.001
S1 EMA xover	0.61	0.29	0.67	0.086

Table 3: Run-length-bootstrap Monte-Carlo Sharpe p -values, $N=1000$. S2/S3/S4 reject random at $p < 0.005$. S1 is only marginal at the 5% level — consistent with the Harris–Yilmaz finding that LF-filter direction out-performs MA crossover.

5.3 Phase 2 – Per-Stock Full History

Single-config full-history backtests on 29 constituents, both base and BIST100-HP regime-gated, run via `src/eval/run_phase2.py`:

	HP base	HP regime	LOWESS base	LOWESS regime
mean Sharpe	0.419	0.372	0.454	0.421
mean CAGR	9.82%	7.56%	11.37%	9.11%
mean MDD	−70.4%	−66.6%	−72.9%	−65.7%
total trades	16,717	11,630	17,226	11,592
Sharpe>0	26/29	28/29	27/29	28/29

Table 4: Phase-2 per-stock summary on 29 redenomination-adjusted BIST100 constituents.

Reading. The regime filter cuts trade count by ~30% (entries gated, exits not), trims mean MDD by 2–5 pp, and lifts the Sharpe-positive count from 26/29 to 28/29 — but it does *not* uniformly raise Sharpe (base edges regime on the mean). It rescues weak tickers (BIMAS, ENKAI, ARCLK) while hurting some strong ones (TKFEN, TOASO, VESTL, HEKTS). The headline interpretation: the regime acts as a *variance reducer*, not a *Sharpe enhancer*.

5.4 Phase 2 – Per-Stock Monte Carlo

Roughly 55% of constituents reject random at 5% under HP; 62% under LOWESS. At the 1% level the counts are 28% and 34% respectively.

5.5 Phase 2 – Equal-Weight Portfolio

The cross-sectional equal-weight portfolio (Section 4.7) is the headline DESIGN.md M4 number:

	HP	LOWESS
$p < 0.01$	8/29	10/29
$p < 0.05$	16/29	18/29
mean strat Sharpe	0.436	0.465

Table 5: Per-stock MC Sharpe p -values, $N=1000$, base variant. Regime variant is intentionally not MC-tested because the gated-entry / sticky-flat logic is path-dependent and would be mis-stated by the fast simulator.

Metric	HP base	HP regime	LOWESS base	LOWESS regime
Sharpe	0.895	0.823	1.049	0.958
Sortino	1.07	0.98	—	—
CAGR	15.29%	11.94%	18.15%	13.74%
Ann. volatility	17.7%	15.1%	17.3%	14.6%
Max drawdown	−26.8%	−25.1%	−28.5%	−25.5%
Calmar	0.57	0.48	—	—
Total return (24y)	29.6×	14.7×	52.9×	21.4×
Mean active tickers	24/29	19/29	24/29	19/29

Table 6: Equal-weight cross-sectional portfolio metrics. The portfolio Sharpe (~ 0.9 for HP, ~ 1.05 for LOWESS) is roughly $2\times$ the mean per-ticker Sharpe (~ 0.42 – 0.46) — the diversification benefit of cross-sectional aggregation.

Reading.

- LOWESS portfolio Sharpe 1.05 exceeds BIST100 buy-and-hold Sharpe 0.92. HP portfolio Sharpe 0.90 is essentially at parity with buy-and-hold.
- Total nominal return is $\sim 22\%$ of buy-and-hold for HP and $\sim 39\%$ for LOWESS — i.e. the strategies give up substantial nominal upside in exchange for $\sim 2\times$ shallower drawdowns (-27% vs -63%).
- The regime variant lowers volatility (by $\sim 2.6\text{pp}$ for HP, $\sim 2.7\text{pp}$ for LOWESS) and tightens MDD marginally, but at the cost of ~ 3 – 4pp of CAGR and ~ 0.07 – 0.09 of Sharpe.

5.6 Phase 2 – Walk-Forward (S3 HP)

A 6-configuration grid ($\lambda \in \{1600, 14400, 129600\} \times W \in \{252, 504\}$), with 3-year train, 1-year test, 1-year step, produces 477 fold rows per variant:

	mean test Sharpe	median	% folds > 0	mean test MDD
Base	0.358	0.369	66.3%	−25.4%
BIST100-HP regime	0.294	0.295	63.7%	−25.6%

Table 7: Walk-forward summary, S3 HP, 29 tickers.

Parameter selection. Across base folds, $W=252$ is picked in 466/476 folds; λ splits 192/141/133 across $\{129600, 14400, 1600\}$. The slow $\lambda \times$ long W combinations are essentially unused — they should be trimmed in any next-iteration grid.

Best / worst tickers. Best by mean fold Sharpe: VESTL 0.86, ULKER 0.72, TKFEN 0.68, HEKTS 0.67, KRDM 0.67, KCHOL 0.66, ISCTR 0.63. Worst: BIMAS -0.28 , ENKAI -0.23 , TCELL 0.00, SAHOL 0.10, ASEL 0.11.

5.7 HP vs LOWESS Head-to-Head

Metric	S3 HP	S4 LOWESS	Δ (L–H)
mean Sharpe (base)	0.419	0.454	+0.035
mean Sharpe (regime)	0.372	0.421	+0.049
mean MDD (base)	-70.4%	-72.9%	-2.5pp
mean MDD (regime)	-66.6%	-65.7%	$+0.9\text{pp}$
total trades (base)	16,717	17,226	+509
Sharpe>0 (base)	26/29	27/29	+1
MC $p < 0.01$	8/29	10/29	+2
MC $p < 0.05$	16/29	18/29	+2
mean MC Sharpe	0.436	0.465	+0.029
Portfolio Sharpe (base)	0.895	1.049	+0.154
Portfolio Sharpe (regime)	0.823	0.958	+0.135
Portfolio CAGR (base)	15.29%	18.15%	+2.86pp
Portfolio CAGR (regime)	11.94%	13.74%	+1.80pp
Portfolio total return (base)	29.6×	52.9×	+78%

Table 8: Head-to-head HP vs LOWESS across every aggregation level.

Verdict. LOWESS dominates HP at every aggregation level — per-stock (+0.04 to +0.06 Sharpe), MC (more tickers significant at both 1% and 5%), and portfolio (+0.15 Sharpe, +2.8pp CAGR, $\sim 80\%$ more total return). The original choice of S3 HP as the primary strategy was the theoretically motivated one (Harris–Yilmaz LF-filter argument); the writeup reports **HP as primary on principle and LOWESS as the better empirical alternative** rather than swapping winners. Both reject random handily; the gap is real but modest in significance terms ($8 \rightarrow 10$ tickers at $p < 0.01$, $16 \rightarrow 18$ at $p < 0.05$).

6 Findings and Discussion

6.1 Summary of Key Findings

1. **Low-frequency filters beat moving-average crossovers**, as predicted by Harris and Yilmaz. On the BIST100 Index, S3 HP and S4 LOWESS both produce Sharpe > 0.8 and reject the matched-random null at $p < 0.005$, while S1 EMA crossover only reaches Sharpe 0.61 and is marginal at $p = 0.086$.
2. **Trend signal direction (S2/S3/S4) is the consistent winner**, suggesting that the act of *differencing the smoother* — rather than *comparing two smoothers* — is where the edge lives.
3. **LOWESS narrowly outperforms HP empirically at every aggregation level** (per-stock, MC, portfolio). The gap is real but modest in statistical terms.
4. **The regime filter is a variance reducer, not a Sharpe enhancer.** Across both HP and LOWESS, gating with the BIST100-HP filter cuts trade count $\sim 30\%$ and trims MDD

by 2–5pp, but reduces mean Sharpe slightly. It rescues the worst-of-base tickers (THYAO, BIMAS, ENKAI) while hurting some of the best (TKFEN, TOASO, VESTL).

5. **Cross-sectional diversification roughly doubles Sharpe.** The ~ 0.42 mean per-ticker Sharpe becomes a ~ 0.90 portfolio Sharpe — the standard diversification dividend from 29 imperfectly correlated bets.
6. **The strategy is a drawdown reducer, not a return enhancer.** Versus BIST100 buy-and-hold (CAGR 23.5%, MDD -63.4%), the LOWESS portfolio (CAGR 18.2%, MDD -28.5%) gives up ~ 5 pp of CAGR for roughly half the drawdown. The HP portfolio gives up ~ 8 pp of CAGR for the same drawdown profile.
7. **Out-of-sample Sharpe materially lower than in-sample.** The 3y/1y/1y walk-forward yields a ~ 0.36 mean test Sharpe per ticker, well below the 0.82 in-sample index Sharpe. This is the honest cost of fold-level re-estimation on noisy individual names rather than the smoother index.
8. **Parameter robustness.** Across walk-forward folds, the short window ($W=252$) wins 466/476 times. λ is much less determined: all three values appear meaningfully (133, 141, 192 selections). The grid should be trimmed in any next iteration.

6.2 The Sharpe vs CAGR Trade-Off Frontier

Approach	Sharpe	CAGR	MDD
BIST100 buy-and-hold	0.92	23.46%	-63.4%
HP portfolio (base)	0.90	15.29%	-26.8%
HP portfolio (regime)	0.82	11.94%	-25.1%
LOWESS portfolio (base)	1.05	18.15%	-28.5%
LOWESS portfolio (regime)	0.96	13.74%	-25.5%

Table 9: Risk-adjusted comparison. Only LOWESS-base produces a Sharpe ratio meaningfully above buy-and-hold; all four portfolio variants cut MDD by ~ 35 – 40 pp.

The natural reading: a leverage-tolerant investor with a moderate $\sim 30\%$ drawdown budget can scale the LOWESS-base portfolio by $\sim 2.3\times$ to match buy-and-hold’s MDD, in which case its expected CAGR would rise to $\sim 42\%$ — well above buy-and-hold’s 23%. (This is a back-of-envelope calculation; an honest scaled version would need a re-run with margin financing costs and short-borrow costs explicitly modelled.)

6.3 Why the Regime Filter Underperforms on Sharpe

Asymmetric gating is the proximate culprit: the filter blocks *entries* but not *exits*. This produces *sticky-flat* stretches where the signal has flipped (in favour of a new direction), but the regime disagrees, so the strategy sits in cash — missing the early part of the new move. When the regime later confirms, the strategy enters late at a worse price. The losses on early-entry days that the filter *would have caught* are typically smaller than the missed early gains on the days the filter blocks.

6.4 Comparison with Harris–Yilmaz (2008)

Harris and Yilmaz study daily FX, not equities. Two systematic differences matter:

- **Drift sign.** Equities have a positive unconditional drift, FX pairs do not. Always-in-market shorts on equities therefore bleed against the drift even on good signals. This explains the large per-stock MDDs and the strategy’s underperformance versus buy-and-hold on absolute return.
- **Volatility regimes.** Turkish equities exhibit the FX-like regime-of-regimes structure (currency crises, 2018 lira crisis, etc.) but with additional equity-specific factors (sector cycles, corporate actions). The HP/LOWESS direction signal captures the regime-of-regimes well; sector-specific factors are uncaptured.

The structural Harris–Yilmaz conclusion — LF-filter direction beats MA crossover — replicates in our setting. The magnitude of the edge is smaller, which is what one would expect when transferring an FX result to an equity universe with positive drift and shortable but costly individual names.

7 Improvement Ideas and Future Work

The following are concrete, prioritised improvement directions. Each is scoped small enough to be tackled as an independent piece of work; none blocks the others.

7.1 Data and Universe

IM1 – Full official BIST100 membership. The current 29-name universe is a curated subset of well-known blue chips. Swapping in the official current 100-name membership (subtracting whichever names still fail yfinance) would treble the cross-sectional sample, improve diversification, and let us report per-sector breakdowns honestly. *Estimated effort:* 1–2 days, mostly ticker remapping for the names that still fail ingest.

IM2 – Point-in-time BIST100 membership (no survivorship). The brief permits survivorship bias, but the most defensible upgrade is to use *historical point-in-time* membership: at each rebalance date, only trade names that were in the index at that time. Bloomberg and Refinitiv ship this; constructing it from public sources is laborious but not infeasible. *Estimated effort:* 1–2 weeks if done from scratch, 1 day if a vendor file is available.

IM3 – Adjusted-close vendor data. The `_adjust_splits(3×)` threshold catches the 2005 redenomination and the MGROS 2009 gap; but a single threshold is fragile. Using a proper corporate-actions feed (vendor adjusted-close with audited split, dividend and rights-issue factors) would eliminate the threshold-tuning fragility entirely.

IM4 – Currency-translated returns. All returns are in TL. The TL has depreciated ~90% in USD terms over the sample. A USD-denominated view would expose to a foreign investor what fraction of the local Sharpe survives FX translation. This is a ~1-day add-on (multiply by USDTRY, compute Sharpe).

7.2 Signal Engineering

IM5 – HP + LOWESS signal ensemble. The two LF-filter signals are correlated but not identical (Phase-1 Sharpe gap of 0.06, per-stock Sharpe gap of 0.04). A simple AND-gate (trade only when both agree) or weighted vote could be tested cheaply. *Hypothesis:* ensembling cuts whipsaws at the cost of a slightly slower entry, with net Sharpe improvement.

IM6 – Adaptive λ . The walk-forward shows λ -selection is unstable across folds (no single λ dominates, unlike $W=252$). A volatility-adaptive $\lambda(t) = \lambda_0/\sigma_t^2$ (more smoothing in high-vol regimes) is a one-line change and would test whether the instability is genuine or an artefact of grid coarseness.

IM7 – Slope magnitude, not just sign. The current rules use only $\text{sign}(\Delta\tau_t)$. Using $\Delta\tau_t/\sigma_t$ as a *continuous* signal that scales position size would give a smoother allocation and might reduce whipsaw losses. This is a meaningful re-design — the brief’s binary-trend convention would need to be relaxed.

IM8 – HP residual as a contrarian overlay. The high-frequency residual $c_t = P_t - \tau_t$ is by construction mean-reverting. A short-term mean-reversion overlay using c_t could be added as an orthogonal alpha; the two should diversify well.

IM9 – Kalman / state-space trend. The HP filter is a special case of a local-linear-trend Kalman model with specific signal-to-noise ratio. Estimating the SNR from data (rather than fixing λ) is a principled upgrade with virtually no extra implementation cost on top of an `statsmodels.UnobservedComponents` call.

7.3 Position Sizing and Risk Management

IM10 – Volatility targeting. Current sizing is fixed-cash (100k TL per order). A natural improvement is to size to a constant *volatility* target: $\text{notional}_t \propto 1/\hat{\sigma}_t$. This is standard in CTA practice; it should reduce realised vol drift and improve risk-adjusted return at the portfolio level. The brief’s `FixedCashSizer` could be replaced with a volatility-targeted sizer without otherwise touching the pipeline.

IM11 – Cross-sectional risk-parity weighting. The portfolio is currently equal-weight. Risk-parity ($\text{weight} \propto 1/\sigma_i$) would reduce the dominance of high-vol names in the realised portfolio variance. This is a one-line change in `src/eval/run_phase2_portfolio.py`.

IM12 – Drawdown stop-out. Per-ticker MDDs of -50% to -90% are the cost of always-in-market shorts on names with positive drift. A per-ticker drawdown stop (e.g. exit and wait for the next signal flip if the rolling-1y drawdown exceeds 40%) would directly attack the worst realised paths. Requires careful handling so it doesn’t degenerate into trend-chasing.

IM13 – Position-level concentration limits. Single-name concentration is currently bounded only by capital. An explicit limit ($\leq 10\%$ of equity per name) would protect against catastrophic single-name events.

7.4 Backtest Realism

IM14 – Transaction costs. The brief mandates zero commissions, which the headline backtest respects. A sensitivity analysis with realistic BIST commissions (typically 5–15 bps round-trip), bid–ask spread (5–20 bps depending on liquidity), and short-borrow cost (highly name-dependent; typically 1–5% annualised for liquid names, much higher for the small-cap short side) would tell us how much of the Sharpe survives. Expected impact: ~ 0.1 – 0.3 Sharpe loss for the high-trade-count strategies.

IM15 – Slippage model. Market orders fill at the next-bar open in Backtrader’s default `Cheat-On-Open=False` mode. A more honest fill model (e.g. a fraction of average daily volume capped at 5% of ADV) would surface any liquidity-constrained names. PGSUS and ARCLK are candidates.

IM16 – Borrow availability. Shorts on Turkish blue chips are generally feasible but not always cheap. A two-tier model (block the short if borrow > X bps, scale down if borrow $\in [Y, X]$) would be the right realism upgrade. Out of scope of the brief but should be acknowledged.

7.5 Statistical Testing

IM17 – Regime-variant per-stock MC. The current fast Monte-Carlo simulator cannot model the regime variant’s sticky-flat behaviour and so the per-stock MC table is base-only. Building `monte_carlo_sharpe_regime(prices, signal, regime, n_iter)` on top of the existing `effective_position` state machine in `src/eval/portfolio.py` is a ~ 50 -line change and would close the methodological gap noted in DESIGN.md §10.4.

IM18 – Portfolio-level Monte Carlo. The portfolio Sharpes (0.90 HP / 1.05 LOWESS) deserve their own significance test against a matched-random *portfolio*. Per MC iteration: draw an independent matched-random signal per ticker, build the effective-position panel, equal-weight-average, compute portfolio Sharpe. With $N = 500$ the resolution at $p < 0.01$ is fine and the runtime is tractable.

IM19 – Stationary block bootstrap. The current run-length match preserves trade-frequency moments but assumes the run-length distribution itself is stationary. Politis and Romano’s stationary block bootstrap is the standard upgrade, blocking on the price series itself rather than the signal.

IM20 – White’s reality check / SPA test. With four strategies and multiple parameter configurations evaluated on the same series, the single-strategy Sharpe p -values are subject to multiple-testing inflation. White’s (2000) reality check and Hansen’s SPA test (2005) are the standard corrections; both have public Python implementations.

7.6 Walk-Forward and Robustness

IM21 – LOWESS walk-forward. The walk-forward driver currently hardcodes `hp_direction_signal`. Adding a `-strategy lowess` arm and running it (~ 25 – 40 minutes on the full universe) would close the HP-vs-LOWESS head-to-head out-of-sample. Recommended for the next iteration; not strictly required by the brief.

IM22 – Trim the WF grid. $W=504$ is essentially never picked (10/476 folds); dropping it halves the WF runtime with no measurable loss.

IM23 – Combinatorial purged cross-validation (Lopez de Prado). Walk-forward is a single time-ordered split, vulnerable to a single bad fold. Purged k -fold CV with embargo is the modern standard for financial time-series CV. Implementation effort: ~ 1 – 2 days.

IM24 – Deflated Sharpe ratio. Lopez de Prado’s *deflated Sharpe ratio* (2014) corrects for both selection bias (we tried 41 Phase-1 configurations) and the higher moments of returns. Should be reported alongside the raw Sharpe in any updated table.

7.7 Engineering and Reproducibility

IM25 – Feature parquet cache. DESIGN.md §5.2 planned a parameter-keyed `data/features/` cache for indicator parquets and explicitly deferred it. With the LOWESS portfolio taking ~25 minutes (vs ~1 minute for HP) because each rolling refit cannot be pre-factored, the cache becomes worthwhile as soon as we extend the LOWESS grid materially.

IM26 – Restore the venv. The repository’s `.venv` was created under the old path `~/ec532-project` (the repo was renamed). Console scripts with hard-coded shebangs (`pip`, `jupyter`, `nbconvert`) fail with *bad interpreter*. Workaround in use: invoke via `.venv/bin/python -m <module>`. A clean rebuild of the venv would resolve this permanently.

IM27 – Git initialisation. The project is not yet a git repository. A `git init` + initial commit would enable per-run result reproducibility hashes in the parquet metadata.

IM28 – Test coverage for causality. The most consequential methodological pitfall in this project is HP/LOWESS look-ahead bias. A unit test that compares `rolling_hp_trend(P, lam, W)[t]` against `hp_filter(P[:t+1], lam)[-1]` would catch any future regression where the wrapper accidentally peeks ahead.

7.8 Reporting Extensions

IM29 – Per-sector slice. `config.py` has placeholder sector comments. Adding a sector tag column and a `groupby` in the portfolio aggregator would give per-sector Sharpe tables (banks, holdings, transport, energy, industrials, etc.). A 1-hour change.

IM30 – Regime-conditional return tables. Decomposing portfolio returns into “BIST100 trending up” vs “BIST100 trending down” weeks would directly visualise where the strategy makes its money. Useful in the deck.

IM31 – Drawdown attribution. For each MDD episode $> 10\%$ in the portfolio curve, attribute the loss to (a) underlying market move, (b) signal-side losses, (c) regime gating. This is the diagnostic-style table that gives a reader confidence in the variance-reducer interpretation.

8 Limitations

1. **Survivorship bias.** The 29-name universe is the *current* membership snapshot. Names that entered then left the index, and names that delisted in the sample, are absent. The brief permits this but a point-in-time membership reconstruction (IM2) is the right next step.
2. **Zero transaction costs.** Mandated by the brief, but the strategies trade 16,000–17,000 round-trips on the 29-ticker universe over 24 years. At a realistic 10 bps round-trip cost, the unhedged Sharpe loss would be ~0.1–0.2 (IM14).
3. **No short-borrow cost.** Equally mandated by the brief; arguably the more important omission given that the strategy spends ~50% of its life short.
4. **No corporate-action vendor feed.** Splits, rights issues and dividends are not properly back-adjusted; we use a single $> 3\times$ price-jump heuristic in `_adjust_splits`. Robust to the 2005 redenomination and the MGROS 2009 gap, but could miss smaller corporate actions (IM3).

5. **Single primary strategy.** S3 HP is the primary on theoretical grounds; LOWESS is the empirical winner. A model-selection upgrade (cross-validated Sharpe on a holdout fold) would resolve the choice on data rather than priors.
6. **Backtrader multiprocessing not used.** The brief warns this optimiser can be flaky; we used sequential loops. With a feature cache (IM25) this becomes a non-issue.
7. **Single-window walk-forward.** The 3y/1y/1y split is one specific choice; combinatorial purged CV (IM23) would test robustness to that choice.
8. **Sharpe-only selection criterion.** Hyperparameter selection uses only Sharpe (with a min-trade floor). Selection by Calmar, Sortino, or a multi-objective rule would likely pick different parameters.

9 Conclusion

We implemented a four-variant trend-following strategy on Turkish equities inspired by Harris and Yilmaz (2008), with full respect for the EC581 course brief’s constraints: Backtrader engine, 1M TL initial capital, 100k TL per-trade `FixedCashSizer`, zero commissions, daily bars, and a matched-random Monte-Carlo p -value with $N=1000$.

The headline empirical findings are:

- **The Harris–Yilmaz hypothesis replicates.** Low-frequency filter direction (S3 HP, S4 LOWESS) beats classical MA crossover (S1) on the BIST100 Index, both in raw Sharpe and in Monte-Carlo significance.
- **HP is the principled primary; LOWESS is the empirical winner.** The gap is small (0.04–0.15 Sharpe across aggregation levels) but consistent; we report both.
- **The cross-sectional portfolio doubles per-ticker Sharpe.** From ~ 0.42 mean per-stock to ~ 0.90 (HP) and ~ 1.05 (LOWESS) portfolio — a textbook diversification dividend.
- **The regime filter is a variance reducer.** It cuts trade count $\sim 30\%$, tightens MDD slightly, but slightly reduces mean Sharpe. It rescues weak tickers and clips strong ones.
- **The strategy gives up nominal return for drawdown reduction.** HP/LOWESS portfolios produce $\sim 22\text{--}39\%$ of buy-and-hold’s total return at $\sim 42\%$ of buy-and-hold’s MDD.

The most impactful improvement directions are: (i) adding transaction-cost and short-borrow realism (IM14–16), (ii) upgrading the universe to point-in-time BIST100 membership (IM2), (iii) implementing the regime-variant and portfolio-level Monte-Carlo (IM17–18), and (iv) moving to volatility-targeted sizing and risk-parity weighting (IM10–11). Each is a 1-day to 1-week piece of work and individually addresses a distinct limitation in the current methodology.

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A Code Structure

```
ec581-project/
|-- DESIGN.md           # design document (full spec)
|-- CLAUDE.md           # development memo
|-- NB_Projects.pdf     # course brief
|-- config.py           # paths, seeds, capital, universe
|-- requirements.txt
|-- data/               # gitignored
|   |-- raw/             # yfinance ingest, immutable
|   |-- clean/           # cleaned, redenomination-adjusted
|   |-- panel/           # wide prices/volumes + bist100 OHLCV
|-- results/            # parquet outputs (sweep, MC, portfolio, WF)
|-- notebooks/
|   |-- 01_eda.ipynb
|   |-- 02_index_strategy_comparison.ipynb
|   |-- 03_walk_forward.ipynb
|   |-- 04_regime_filter.ipynb
'-- src/
    |-- data/            # ingest, clean, panel, build
    |-- features/        # ema, hp, lowess, regime (causal wrappers)
    |-- backtest/        # sizers, feeds, strategies, runner, sweep
    '-- eval/            # metrics, montecarlo, walkforward,
                        # portfolio, strategies registry,
                        # CLIs: run_phase2[_mc|_portfolio],
                        #         run_walkforward
```

B Reproducibility

1. Build data: `.venv/bin/python -m src.data.build` (full universe) or `.venv/bin/python -m src.data.build -smoke` (M1 subset).
2. Phase-1 sweep: `.venv/bin/python -m src.backtest.sweep` → `results/phase1_sweep.parquet`.
3. Phase-1 Monte Carlo: `.venv/bin/python -m src.eval.montecarlo` → `results/phase1_mc.parquet`.
4. Phase-2 per stock: `.venv/bin/python -m src.eval.run_phase2 -strategy {hp,lowess}`.
5. Phase-2 per-stock MC: `.venv/bin/python -m src.eval.run_phase2_mc -strategy {hp,lowess} -n-iter 1000`.
6. Phase-2 portfolio: `.venv/bin/python -m src.eval.run_phase2_portfolio -strategy {hp,lowess}`.
7. Phase-2 walk-forward: `.venv/bin/python -m src.eval.run_walkforward` (HP only as of this writing).
8. Notebook re-execution: `.venv/bin/python -m jupyter nbconvert -to notebook -execute notebooks/<nb>.ipynb -output <nb>.ipynb`.

All MC and bootstrap seeds are sourced from `config.RANDOM_SEED=42`.